# Distributed Multi-Agent Deep Reinforcement Learning based Navigation and Control of UAV Swarm for Wildfire monitoring

Adeeba Ali
*Department of Computer Engineering*
*Aligarh Muslim University*
Aligarh, India
aliadeeba98@gmail.com

Rashid Ali
*Department of Computer Engineering*
*Aligarh Muslim University*
Aligarh, India
rashidaliamu@rediffmail.com

M.F. Baig
*Department of Mechanical Engineering*
*Aligarh Muslim University*
Aligarh, India
drmfbaig@yahoo.co.uk

2023 IEEE 4th Annual Flagship India Council International Subsections Conference (INDISCON) | 979-8-3503-3355-8/23/$31.00 ©2023 IEEE | DOI: 10.1109/INDISCON58499.2023.10270198

***Abstract*—Decentralized deep reinforcement learning is an emerging and most effective approach to solve the problem of resource allocation and coordination among the swarm of UAVs. Nowadays, the use of autonomous aerial vehicles in wildfire monitoring is increasing and considered as the reasonably feasible option as surveillance in calamity-hit areas can benefit from this kind of automation. The flocks of UAVs can generate maps of affected areas which could improve the process of relief planning so that necessary aid can be reached the burnt areas quickly. This paper presents the Multi-agent Deep Q network-based technique for planning optimized trajectories for the UAV swarm which can sense the wildfire in the forests and nearby regions. In this work, the UAV agents are trained over simulated wildfires in virtually generated forests with two reward schemes. The simulation results verify the effectiveness of the proposed strategy for leveraging it in real-world scenarios.**

***Index Terms*—swarms, reinforcement learning, navigation, wildfire, decentralized control, UAV**

## I. Introduction

Since the last few years, the spread of massive wildfires all across the world has caused substantial environmental, economic, and social effects [1]. And, the most dreadful effects of wildfire include the death of human beings, as well as other medical problems that arise due to high temperature, smoke inhalation, debris, and vehicle crashes. Similarly, the adverse effects of wildfires on environmental health include degradation of air quality and the change in the overall global composition of the atmosphere as wildfires emit Carbon Dioxide (CO2) in large quantities, thus leading to global warming. Furthermore, uncontrolled wildfires are also responsible for the extinction of some species of flora and fauna in that particular location.

In order to reduce the amount of loss caused due to wildfires, it is important to keep account of the information regarding the fuel map, movement, and spread of fire front and climate conditions. The availability of this kind of accurate real-time information can reduce the risk concerned with the lives of firefighters and increase success in the efforts made to control the spread of fire [2]. The images captured by the satellites are one of the key ways to gather information about wildfires in real-time. However, these images can only be made available if the satellite goes over that particular region of interest. Additionally, satellite photos with tiny flames have low spatial resolution [3]. Man-made aerial aircraft, which are rather expensive and need a highly competent human pilot, are the other alternate option for getting useful information. One of the most interesting alternatives is the use of autonomous UAVs (Unmanned Aerial Vehicles) to monitor wildfires and record the information in real-time [3]. In addition to the commands transmitted via remote controller, the flight controller used in the autonomous aerial vehicles can also accept commands guiding the exterior maneuver. The GNSS coordinates and IMU (Inertial Measurement Unit) state data are included in the external commands. External commands may be sent directly from the processor incorporated in the drone hardware and physically attached to the flight controller, or they may be transmitted from a ground workstation via a communication medium. However, in some nations, it is needed to designate at least one safety pilot to manually send commands to the drone in emergency situations, despite the fact that autonomous UAVs often only require a single operator to launch the system.

The use of multiple UAV agents instead of a single UAV increases the robustness and efficiency of the system [2]. Furthermore, the use of multiple smaller and cheaper aerial vehicles would minimize the overall cost of the system, thereby increasing the adoption of this kind of system by organizations that possess comparatively fewer financial resources. In contrast, some additional challenges are associated with the use of multiple UAVs, such as the need for an efficient algorithm that can establish proper coordination among the UAVs. Although the UAVs equipped with lasers and infrared sensors are barely used by all public and private services for wildfire detection due to high operational costs, hence the autonomous cooperative control of UAV swarms especially for the task of wildfire monitoring is yet to be further investigated [4-9]. In the literature, collaborative control of multiple UAV agents can be achieved either by planning optimized and predefined trajectories to the goal position or using techniques

979-8-3503-3355-8/23/$31.00 ©2023 IEEE

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on April 07,2026 at 11:43:23 UTC from IEEE Xplore. Restrictions apply.

based on decentralized control theory or RHO (Receding Horizon Optimization). In [11] the authors proposed an approach that outperforms the traditional RHO method for wildfire monitoring with two aircraft. The technique presented in [11] is based on deep reinforcement learning (Deep-RL), and the results demonstrated in the paper shows that learning-based mechanisms for cooperative control tasks are promising alternatives. Typically, the researchers leverage off-policy Q-learning [12-13] which is an uncomplicated and classical approach in the field of navigation and control of multiple robots. In [5] deep reinforcement learning has been previously used for autonomous wildfire monitoring through multiple UAVs.

The objective behind this research is to investigate and validate the techniques that can efficiently deal with the challenges associated with collaborative multi-agent reinforcement learning leveraged for surveillancing a wildfire with multiple UAVs. Inspired by [5] we propose a DQN (Deep Q-Networks) based solution for cooperative control of multiple UAVs while monitoring the wildfire. In addition to this, the effect of reward sharing is explored in this research, which is considered as another possible feedback scheme in swarm applications [14].

The rest of the paper is organized as follows. The review of related work is presented in Section II. Then the problem is formally stated in Section III. Next, the implementation details of the formulated problem as POMDP (Partially Observable Markov Decision Process) are discussed in Section IV. This is followed in Section V by a detailed description of the proposed solution for wildfire monitoring. Then the proposed methodology is validated through simulations in Section IV. Finally, the summary of the paper is presented in Section VII.

## II. Related Work

UAVs are used to gather valuable data and information through surveillance and continuous monitoring in a wide range of applications, including smart agriculture [15], post-disaster hazardous material [16], or wildfire monitoring [2].

In this work, the use of multiple UAVs in wildfire monitoring applications [2] is studied. Earlier, wildfire detection, fire front tracking, and hotspot hunting after fire have all been accomplished using thermal imagers and visible light cameras, as well as a combination of both [3]. [17] provides an overview of various infrared-light-based methods for monitoring wildfires from UAVs and satellites.

In contrast to satellite imaging, aerial wildfire detection, and monitoring provide a high-quality temporal and spatial resolution [3], [18]. The authors of [19] illustrated how a remotely controlled UAV may be used to monitor wildfires in sizable regions. In comparison to using a single UAV, using many UAVs for surveillance and monitoring is more effective [3]. Techniques to control the flock of UAVs during the simulation of wildfire detection and monitoring were presented in [2], [7-8], [20-22]. In [23], the authors demonstrated a decentralized mechanism to estimate the wildfire model. The proposed strategy was tested using real-time drone hardware and a stochastic fire simulator was used for generating data. Later, in [24] the authors validated their proposed cooperative control algorithm in real-time field experiments.

The aforementioned methods use principles from control theory to predict the movements of the UAVs. In [11], the authors demonstrated that deep reinforcement learning methods outperformed classical control theory techniques for the task of monitoring wildfires with two airplanes using a random wildfire simulator.

In [6] the authors put forward a deep reinforcement learning strategy to search for burning trees and then utilize a retardant to put out the fire. Similar to [25], the authors looked into an equilibrium-based Q-learning variation to finish the coverage challenge successfully. Moreover, deep reinforcement learning has also been utilized to gather relevant information through learning a particular task. In [27], the authors present the multi-robot information-gathering method (Deep-IG) to allow the agents to capture valuable and relevant information without colliding with each other. With the use of three UAVs, the suggested technique was tested for terrain mapping.

The algorithms proposed in [11], [25-27] were leveraged for learning and establishing coordination among multi-robot agents, thus exemplifying the challenges associated with collaborative multi-agent reinforcement learning problems.

## III. Problem Formulation

### A. Environment

The simulated wildfire surveillance environment is represented through a 2-D map enclosing a 1 $Km^2$ area which is discretized into $100 \times 100$ grid of burnable cells. The generated dynamic maps of the environment are of two types:

- *B(s)*: Fuel map in which each cell represents the amount of flammable material present in that particular region.
- *F(s)*: Fire map which is a boolean representation of the environment consisting of either the value '0' or '1' in each cell where '1' represents the presence of fire in that particular cell.

The simulated wildfire environment with the evolution of fire across the maps over an episode is shown in Fig. 1. The cells of the fuel map are updated every 2.5 sec in accordance to the following two equations:

$$B_{\mathrm{t+1}}(s) = \begin{cases} max(0, B_{\mathrm{t}}(s) - \beta p(s)), \text{if } F_{\mathrm{t}}(s) = 1 \\ B_{\mathrm{t}}(s), \quad \text{otherwise} \end{cases} \tag{1}$$

where $\beta$ is the burning rate which is set to 1, and

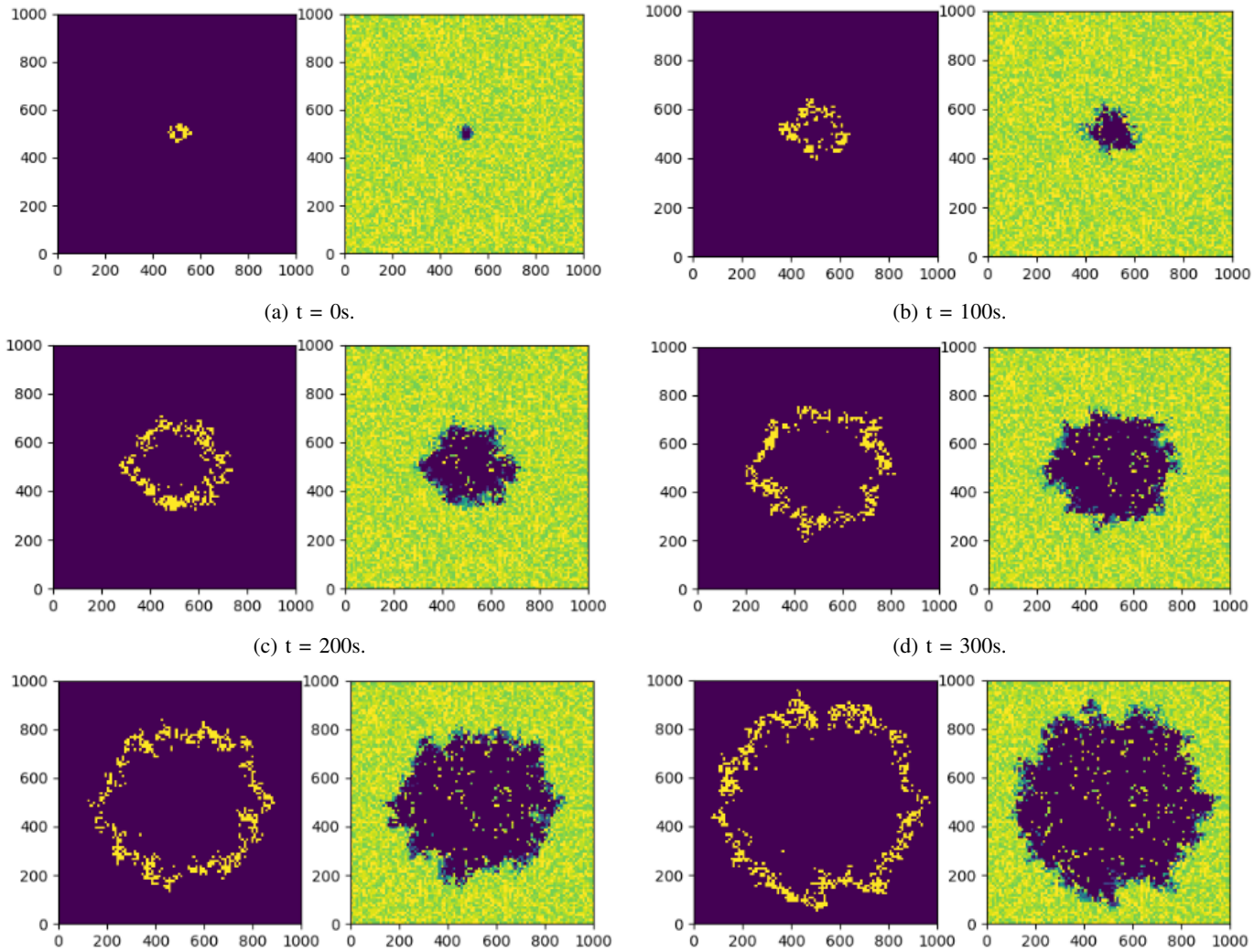
$$p(s) = \begin{cases} 1 - \prod_{s'|d(s,s')<d_{mp}} (1 - P(s,s')F_t(s')), \text{if } B_t(s) > 0 \\ 0, \quad \text{otherwise} \end{cases} \tag{2}$$

where $p(s)$ is the probability that the cell $s$ of the environment will catch fire, $d(s, s')$ is the distance between cells s and s', $d_{mp}$ is the maximum distance at which propagation can happen and $P(s, s')$ is the probability that cell $s'$ ignites cell $s$ which is defined as follows:

$$P(s, s') = w.d(s, s')^{-2} \tag{3}$$

where $w$ is constant proportional to the wind speed.

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on April 07,2026 at 11:43:23 UTC from IEEE Xplore. Restrictions apply.



(a) t = 0s. (b) t = 100s.

(c) t = 200s. (d) t = 300s.

(e) t = 400s. (f) t = 500s.

Fig. 1. An example wildfire evolution over time. Left: burning cells. Right: fuel left.

### B. Agents

Fixed-wing aircraft are considered as agents for this problem. It is assumed that the aircraft will fly with a constant velocity $v$ of 20 ms$^{-1}$ and each agent is allowed to decide the increment or decrement of its bank angle ($\phi$), also known as angle of tilt, by 5 degrees at the 10 Hz frequency. The position of the aircraft is updated according to the kinematic model of a fixed-wing aerial vehicle. The maximum bank angle that a UAV can take is fixed to 50 degrees and UAV maneuvers going beyond this range will not be considered valid. This constraint will set angular velocity within realistic limits.

## IV. SWARMDP DESCRIPTION

Rather than using the "Partially Observable Markov Decision Process" (POMDP) as a model [28], the navigation problem is framed as a special case of Decentralized POMDP, that is swarMDP [29]. Through this design, the multi-agent settings can be explicitly modeled by assuming that all the agents have similar configurations.

**Definition 1.** *The swarMDP problem is formulated in two parts. First, a vector $\dot{\mathbb{A}} = \langle S, A, B, \pi \rangle$, where $\dot{\mathbb{A}}$ represents the local attributes associated with each swarm agent and the value of each tuple variable is determined as follows:*

- *S: collection of states.*
- *A: set of actions.*
- *B: collection of observations.*
- *$\pi$: policy ($B \rightarrow A$).*

*In the second step, the swarMDP is finally defined using a tuple of 7 dimensions $\langle N, \dot{\mathbb{A}}, P, R, O, \gamma \rangle$ where:*

- *N is the ordered collection of indices corresponding to the aircraft. The index values in the set should be in the range from 1 to N, where N represents the total number of UAV agents.*
- *$\dot{\mathbb{A}}$ is the prototype of the agent as defined above.*

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on April 07,2026 at 11:43:23 UTC from IEEE Xplore. Restrictions apply.

- $P : S^2 \times A \rightarrow [0,1]$ *is a function to compute the probability of going from one state to another, where* $P(s'|s,a)$ *denotes the transition probability that the agent will move to state s' from state s after taking action a.*
- $R : S \times A \rightarrow \mathbb{R}$ *is the reward function, where* $R(s,a)$ *is the reward given to an agent when it takes action* $a$ *in state* $s$.
- $O : S \times Z \times A \rightarrow [0,1]$ *is the observation model, where* $O(z'|s,a)$ *is the probability that an agent observes z' after playing an action a in state s.*
- $\gamma \in [0,1]$ *is the discount factor for the total reward accumulated by the UAV agent in the future.*

### A. UAV State

The localization of each aircraft in the local environment depends on the following factors:

- Fuel map $B(s)$ which has to be updated every 2.5 sec by the wildfire simulator.
- The positional coordinates $(x_i, y_i)$.
- The angle of yaw $(\psi_i)$.
- The angle of tilt $(\phi_i)$.

### B. UAV Actions

For each agent the vector of UAV local actions is given by $A = \langle$ decrease angle of tilt $(\phi)$ by 5 degrees, increase angle of tilt $(\phi)$ by 5 degrees $\rangle$. Each agent in the swarm has to select any of these actions at a period of 0.1 sec.

### C. Observations

Every agent in the swarm receives two different types of data: a feature vector that shows some information about the UAV's and other UAVs' nearby conditions, and a picture that shows a partial view of the wildfire as seen by the UAV.

With each agent $i$, a feature vector is associated which is computed after the concatenation of four other vectors related to the local state variables of UAVs: the set of all the bank angles $\phi = \{\phi_k|$k ranges from 1 to N$\}$; the range of distance to other UAVs $\rho_i = \{\rho_{ik}|$k ranges from 1 to N and k should not be equal to i$\}$; the relative heading angle to other UAVs, $\theta_i = \{\theta_{ik}|$k ranges from 1 to N and k should not be equal to i$\}$; the relative heading angle of other UAVs, $\psi_i = \{\psi_{ik}|$k ranges from 1 to N and k should not be equal to i$\}$.

The optical hemispherical camera of the UAV is used to take the photos, which have a maximum range of 500 m and a view angle of 160 degrees pointing downhill. After processing the image is downsampled to the resolution of $30 \times 40$ pixels representing the burning trees and forest.

The image is captured after sampling the points of interest at the horizontal angle of 40 degrees from the cardinal and at 30 degrees of elevation from the lowest point to 10 degrees below the skyline. This strategy results in a high-quality image of the region right below the UAV. The characteristics of this image determine the degree of partial observability of the environment.

### D. Rewards

The reward received by each agent $i$ is the sum of four independent components:

$$r_1 = -\lambda_1 \min_{\{s \in S | F_t(s)\}} d_s, \tag{4}$$

$$r_2 = -\lambda_2 \sum_{\{s \in S | d_s \leq r_0\}} 1 - F_t(s), \tag{5}$$

$$r_3 = -\lambda_3 \phi_0^2, \tag{6}$$

$$r_4 = -\sum_{\{k \in 1,2,\ldots,N \wedge k \neq i\}} \lambda_4 exp(-\frac{\rho_{ik}}{\epsilon}) \tag{7}$$

where $d_s$ is the measure of displacement from the UAV to a given cell $s$ of the map. The aforementioned equations mention negative rewards or penalties for the following:

- Distance from fire $(r_1)$: proportional to the distance of the closest cell set on fire.
- Safe cells nearby $(r_2)$: proportional to the count of safe cells that doesn't ignite in a radius $r_0$.
- High bank angles $(r_3)$: depends on the square of the UAV bank angle $(\phi)$.
- Nearness to other UAV $(r_4)$: sum of work done by all other agents, which, at the point of saturation, equals the constant value of $\lambda_4$.

The constant parameters used in the above equations have been given the following values: $\lambda_1 = 10, \lambda_2 = 1, \lambda_3 = 100, \lambda_4 = 1000, r_0 = 10, c = 100$.

The reason behind tuning the parameters to these values is that going away from the burning region should be penalized heavily and, the negative rewards obtained from the bank angle and getting close to other UAVs should become dominant.

## V. Methodology

### A. Deep Reinforcement learning

The objective behind the presented approach is to determine the policy $\pi$ that can optimally map observations to actions. The reward function R, which was defined in the preceding section, determines how ideal the situation is. The value $Q(s,a)$ for each state-action pair can be calculated from the optimal policy, $\pi$. When an agent, $a$, acts in a state $s$, the expected cumulative reward is represented by the value function $Q(s,a)$. The following Bellman equation can be used to calculate the value of the Q function:

$$Q(s,a) = r(s,a) + \{\gamma \sum_{s' \in S} p(s'|s,a) \max_{a' \in A} Q(s',a')\} \tag{8}$$

For a given Q value function the optimal policy can be deduced as:

$$\pi(s) = \arg\max_{a \in A} Q(s,a) \tag{9}$$

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on April 07,2026 at 11:43:23 UTC from IEEE Xplore. Restrictions apply.

A deep neural network that can approximate the function can be trained to estimate the Q-value using DQN [30, 31]. The network is trained using the loss function defined as follows:

$$L_i(\theta_i) = \mathbb{E}_{s,a,r,s'}[e^2] \tag{10}$$

where e is the Bellman error:

$$e = r(s,a) + \{\gamma \max_{a'} Q(s',a'|\theta^-) - Q(s,a|\theta)\} \tag{11}$$

The agent must make a choice between exploring the state-action space using stochastic policy and using the prior information that was gleaned from the estimation of the optimal policy in earlier states while the network is being trained. The most common method for solving this issue is to use exploratory policy during the initial training iterations and then gradually increase the percentage of activities involving exploitation policy.

Deep Q Networks (DQN) circumvent the inherent instability problem of deep reinforcement learning through two mechanisms:

- By randomizing the training samples using the experience replay method to reduce the correlation effect of the sequence. This can be implemented using a replay memory $\mathbb{E}$.
- By simultaneously training the second (target) network (with parameters $\theta^-$) to predict $Q(s',a')$ while training it at a slower update rate than the former (first) network (with parameters $\theta$), the first network can learn the fixed target.

### B. Network Architecture

In the implemented method the deep Q networks deal with two different types of observations separately. The feature vector $\langle N, \dot{\mathbb{A}}, P, R, O, \gamma \rangle$ goes through a series of dense hidden layers each consisting of 100 neurons with ReLU activation function. Following that, a series of dense convolutional and max pooling layers downsampled the monocular image with incomplete environmental information before feeding it to the dense hidden layers of neurons. Two extra thick layers combine and analyze the outputs from both networks before estimating the Q-values for the agents. The complete network architecture is represented in Fig. 2.

### C. Reward Schemes

During the training of Deep Q networks, two different kinds of reward approaches are compared according to their degree of decentralization. Once the network is trained both the models can be used for distributed control.

1) Independent rewards: Each agent obtains its own reward separately from the environment.
2) Shared reward: It is calculated as the mean of the independent rewards and shared by all the agents.

## VI. Experimental Results

### A. Simulation Setup

Both the DQN-based fire detection models trained with two different reward configurations are tested in simulated environments. Training of the network is performed every 100 steps after the collection of 100 new samples. A small batch size of 2000 samples is used in every iteration of training. These training samples are arbitrarily taken from the last 100,000 examples buffer. The target network is updated after every update of 10 action values, that is, once every 1000 steps. In order to prevent the overfitting of the network with the first sequence of samples, the learning of the network starts after the first 2000 steps. The choice of training the network with exploratory actions starts at a fraction of 1 and then linearly decreases during the initial 80 percent of training time and finally stays at the constant value of 0.1 until the training ends. The simulation experiments have a duration of 10,000 steps (20 minutes). Adam [32] is used as the stochastic optimizer of the network and the learning rate is set to the value of $6 \times 10^{-9}$.

### B. Results

The major evaluation metric of the proposed approach is the accumulated reward of all simulated episodes which is determined as the sum of the instantaneous rewards earned by all agents over an episode. Fig. 3 represents the simulation of an episode demonstrating the behavior learned by the agents. As the wildfire propagates in the region, the UAVs learn to head toward the burning areas and start making wider circles. The third component $(r_3)$ of the reward function, which is the penalty for the collision between aircraft is effective in making the agents fly around the burning region in the same direction and at opposite endpoints of the fire while staying far away from each other. In the small region of interest (ROI), the penalty arises due to the nearness between the UAVs outweighing the other and one of the UAVs stays there making circles of small radii. Furthermore, As demonstrated in Fig. 4 with both the reward configurations agents are allowed to learn the monitoring task effectively by minimizing their penalties on average. It is expected that similar performance is obtained with both the reward schemes for this problem as three of the four reward components $(r_1, r_2, r_3)$ are independent of the learning behavior of other UAV agents that is, only better estimation of the local state of the UAV can affect the margin of improvement.

## VII. Conclusion

In this paper, a DQN-based approach is proposed to provide decentralized control to multiple UAVs for monitoring the spread of wildfires. The deep Q networks take a processed monocular image consisting of partial observation of the surroundings and the local state vectors of the UAV as input and generate the control actions for the agents. The proposed method is tested with 2 reward configurations which differ in their degree of federating multiple UAV agents. Simulation results show that UAVs can effectively monitor the wildfire in a

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on April 07,2026 at 11:43:23 UTC from IEEE Xplore. Restrictions apply.

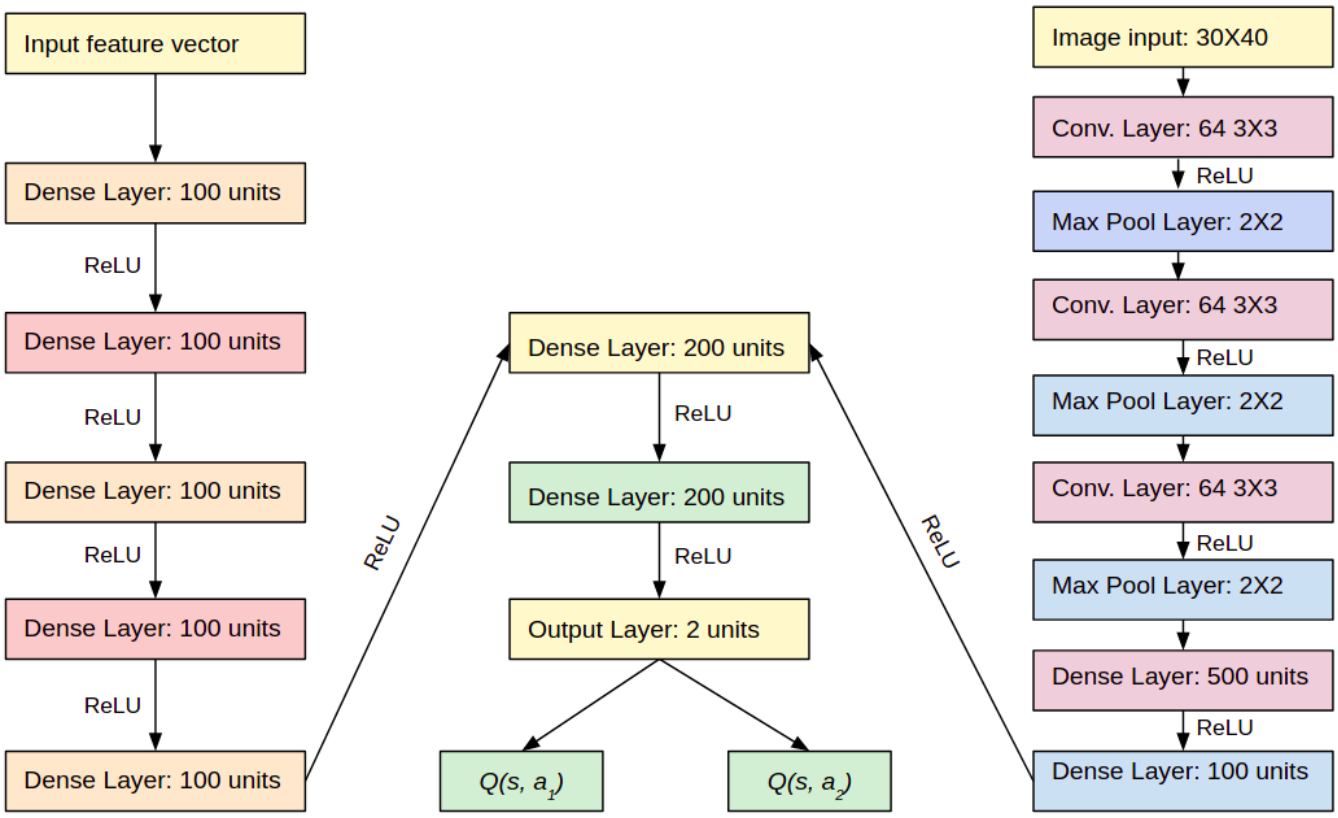



Fig. 2. Network Architecture

cooperative and coordinated manner with both types of reward schemes. Thus, the presented control design can be applied to real-time UAV-aided surveillance in wildfire regions to minimize operational costs. The DQN are trained on simulated fire maps from forests but the model should perform well on all types of wildfires as long as enough data is available for training.

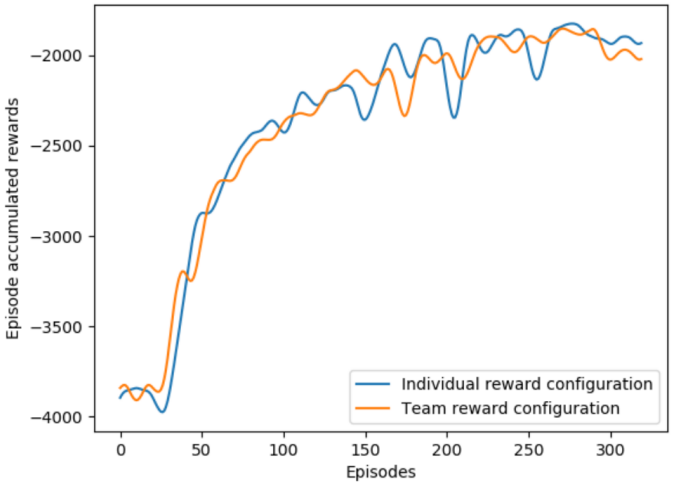



Fig. 4: Comparison of reward schemes at the time of training

## References


[1] S. Doerr, and C. Santin, "Global trends in wildfire and its impacts: Perceptions versus realities in a changing world," Philosophical Transactions of The Royal Society B Biological Sciences, vol. 23, no. 2, pp. 438–454, 2016.

[2] S. Sudhakar et al., "Unmanned Aerial Vehicle (UAV) based Forest Fire Detection and monitoring for reducing false alarms in forest-fires," Computer Communications, Elsevier, 2020, pp. 6648–6653.

[3] F. Saffre et al., "Monitoring and Cordoning Wildfires with an Autonomous Swarm of Unmanned Aerial Vehicles," Drones, MDPI, vol. 6, no. 10, p. 301, 2022.

[4] F. Afghah, A. Razi, J. Chakareski, and J. Ashdown, "Wildfire Monitoring in Remote Areas using Autonomous Unmanned Aerial Vehicles," in MiSARN 2019: Mission-Oriented Wireless Sensor, UAV and Robot Networking, IEEE Infocom Workshops, 2019.

[5] S. Zhang, W. Wang, Z. Liu, and Z. Wu, "Forest Farm Fire Drone Monitoring System Based on Deep Learning and Unmanned Aerial Vehicle Imagery," Mathematical Problems in Engineering, Hindawi, pp. 5181–5188, 2021.

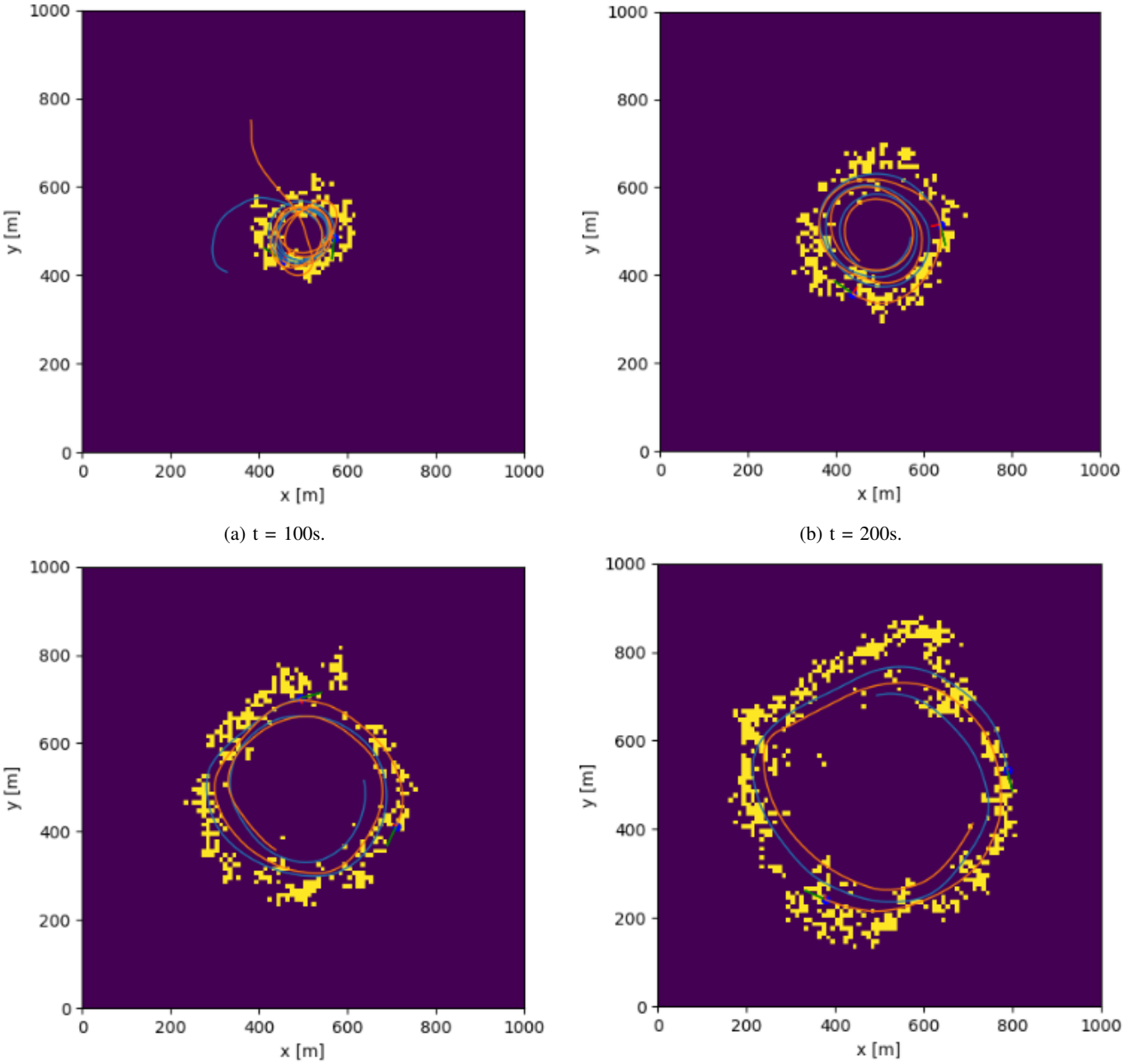
[6] A. Sharma, and P.K. Singh, "UAV-based framework for effective data analysis of forest fire detection using 5G networks: An effective approach towards smart cities solutions," International Journal of Communication Systems, 2021, pp. 1067 1074.

[7] Y. Liu, "Forest Fire Monitoring Method Based on UAV Visual and Infrared Image Fusion," Remote Sensing, MDPI, vol. 39, no. 4, pp. 1530–1548, 2023.

[8] R, N. Haksar and M. Schwager, "Distributed Deep Reinforcement Learning for Fighting Forest Fires with a Network of Aerial Robots," in 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 835–840, 2019.

[9] P. Jain et al., "A review of machine learning applications in wildfire science and management," Canadian Science Publishing, vol. 28, no. 4, pp. 229–255, 2020.


Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on April 07,2026 at 11:43:23 UTC from IEEE Xplore. Restrictions apply.




(a) t = 100s.

(b) t = 200s.

(c) t = 300s.

(d) t = 400s.

Fig. 3. An example of an episode in wildfires with two aircraft trained with DQN, team reward scheme.


[10] M.M. Shahzad et al., "A Review of Swarm Robotics in a NutShell," Drones, MDPI, vol. 7, no. 269, 2023.

[11] X. Mo, D. Peters, and C. Lel, "Low-Cost Autonomous UAV Swarm Application in Wildfire Surveillance and Suppression," in ICMLT '21: Proceedings of the 2021 6th International Conference on Machine Learning Technologies, ACM, 2021.

[12] G. Lozengues, "On the Distributivity of Multi-agent Markov Decision Processes for Mobile-Robotics," International Symposium on Swarm Behavior and Bio-Inspired Robotics, 2021.

[13] A. Oroojlooy, and D. Hajinezhad, "A review of cooperative multi-agent deep reinforcement learning," Applied Intelligence, Springer, vol. 53, pp. 13677–13722, 2023.

[14] X. Zhu, F. Zhang, and H. Li, "Swarm Deep Reinforcement Learning for Robotic Manipulation," Procedia Computer Science, Elsevier, vol. 198, pp. 472-479, 2022.

[15] N. Maurya et al., "Recent Advancement and Role of Drones in Forest Monitoring," Advances in Remote Sensing for Forest Monitoring, 2022.

[16] P. G. Martin, S. Kwong, N. Smith, Y. Yamashiki, O. D. Payton, F. Russell Pavier, J. S. Fardoulis, D. Richards, and T. B. Scott, "3d unmanned aerial vehicle radiation mapping for assessing contaminant distribution and mobility," International journal of applied earth observation and geoinformation, vol. 52, pp. 12–19, 2016.

[17] F. Vivo, M. Battipede, and E. Johnson, "Infra-red line camera data-driven edge detector in UAV forest fire monitoring," Aerospace Science and Technology, Elsevier, vol. 111, p. 106574, 2021.

[18] C. Maffei, R. Lidenbergh, M. Meneti, "Combining multi-spectral and


Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on April 07,2026 at 11:43:23 UTC from IEEE Xplore. Restrictions apply.

thermal remote sensing to predict forest fire characteristics," ISPRS Journal of Photogrammetry and Remote Sensing, vol. 181, no. 8, pp. 400-412, 2021.

[19] C. Hyun, M. Park, W. Lee, "Remotely Piloted Aircraft System (RPAS)-Based Wildlife Detection: A Review and Case Studies in Maritime Antarctica," Animals, MDPI, vol. 10, no. 12, p. 2387, 2020.

[20] V. Sherstjuk, M. Zharikova, and I. Sokol, "Forest Fire Monitoring System Based on UAV Team, Remote Sensing, and Image Processing," in Proceedings of the 2018 IEEE Second International Conference on Data Stream Mining and Processing (DSMP), 2018.

[21] J. Hu et al., "Fault-tolerant cooperative navigation of networked UAV swarms for forest fire monitoring," Aerospace Science and Technology, Elsevier, vol. 123, p. 107494, 2022.

[22] S. Codgay and G. Secinti, "Phoenix: Aerial Monitoring for Fighting Wildfires," Drones, MDPI, vol. 7, no. 1, p. 19, 2023.

[23] I. Alsammaket et al., "Nature-Inspired Drone Swarming for Wildfires Suppression Considering Distributed Fire Spots and Energy Consumption," IEEE Access, vol. 11, 2023.

[24] L. Merino, F. Caballero, J. R. Martínez-de Dios, J. Ferruz, and A. Ollero, "A cooperative perception system for multiple uavs: Application to automatic detection of forest fires," Journal of Field Robotics, vol. 23, no. 3-4, pp. 165–184, 2006.

[25] Z. Mou et al., "Three-Dimensional Area Coverage with UAV Swarm based on Deep Reinforcement Learning," in Proceedings of ICC 2021 - IEEE International Conference on Communications, 2021.

[26] A.Tan, F. Bejarano, Y. Zhu, R. Ren, and G. Nejat, "Deep Reinforcement Learning for Decentralized Multi-Robot Exploration With Macro Actions," IEEE Robotics and Automation Letters, vol. 8, no. 1, pp. 272-279, 2023.

[27] A. Viseras and R. Garcia, "DeepIG: Multi-robot information gathering with deep reinforcement learning," IEEE Robotics and Automation Letters, vol. 4, no. 3, pp. 3059–3066, 2019.

[28] A. Viseras, M. Meissner, and J. Marchal, "Distributed wildfire surveillance with autonomous aircraft using deep reinforcement learning," IEEE Access, vol. 4, pp. 1–11, 2016.

[29] K. Kim et al., "An Application of Inverse Reinforcement Learning to Estimate Interference in Drone Swarms," Entropy, MDPI, vol. 24, no. 10, pp. 1413–1421, 2022.

[30] V. Mnih, K. Kavukcuoglu, D. Silver, A. Graves, I. Antonoglou, D. Wierstra, and M. Riedmiller, "Playing Atari with deep reinforcement learning," arXiv preprint arXiv:1312.5602, 2013.

[31] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski et al., "Human-level control through deep reinforcement learning," Nature, vol. 518, no. 7540, p. 529, 2015.

[32] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," arXiv preprint arXiv:1412.6980, 2014.

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on April 07,2026 at 11:43:23 UTC from IEEE Xplore. Restrictions apply.